

Supplementary Material for: A Survey of Symbol Name Recovery in Software Reverse Engineering

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5 Evaluation Methods

5.1 Datasets

Table 5. Literature with **Unavailable** Data Set and Metrics for Evaluation

Literature	Dataset					Opt	Metrics
	Data Source	# Projs	# Files	# Funcs	Architecture		
UNSTRIP[14]	Glibc datasets, with different gcc and glibc versions.	13	-	-	-	O0-O3	P,R
DREAM++[34]	Example decompiled code from the three decompilers.	-	-	-	-	-	O
Debin[11]	Linux Packages, built with default compilation settings.	830	3.0K	237.0K	x86,x64,arm	O0-O3	P,R,F1
Punstrip[23]	1. Linux packages, compiled with gcc and clang.	5	149	-	x64	O0-O2	P,R,F1
	2. Debian packages.	14.0K	17.5K	-	x64	O0-O2	
Jaffe et al.[15, 16]	Github projects, compiled with gcc.	20.2K	174.4K	-	-	O0	A
Lacomis et al.[19]	Github projects, compiled with gcc.	20.0K	-	-	-	O0	A
SEAM[20]	1. Code generated by cfile, compiled with gcc 6.5.0.	-	-	1.6M	x64	O0	P,O
	2. Github algorithm code, compiled with gcc6.5.0.	-	-	20.0K	x64	O0	
Cfg2Vec[35]	ALLSTAR binaries with more then one arch.	1.1K	27.6K	1.4M	amd,arm,i386,mips	-	P
LmPa[33]	Randomly selected projects, compiled with Clang 12.	6	-	1.3K	-	O0	P,R,O
A2N[32]	Combination of Nero and Coreutils binary files.	-	753.5K	-	-	-	P,R,F1
BinAdapter[21]	AsmDepictor dataset, BinKit[17], and SPEC2006[9].	-	4.3K	760.5K	arm,x64,mips	O2	P,R,F1,O
LLASM[27]	Combination of BinaryCorp [31] and Debin datasets.	-	69.5K	1.6M	-	-	P,R,F1
BETAC[5, 6]	1. Juliet Test Suite and NDSS18, compiled with gcc.	-	97.5K	98.4K	x86	-	A,R,P,O
	2. Ubuntu packages.	50.0K	25.3K	25.4K	amd64	-	
	3. programs with three obfuscation techniques.	-	-	3.7K	-	-	

(Continued)

Table 5. Continued

Literature	Dataset					Opt	Metrics
	Data Source	# Projs	# Files	# Funcs	Architecture		
ADAM[8]	No evaluation.	-	-	-	-	-	-
Anti-ProGuard[2]	F-Droid apps.	-	-	-	-	-	A
Isobe et al.[12, 13]	Maven central repository (MCR) [24] programs.	-	17.7K	935.8K	-	-	P,R
Otsuki et al.[22]	No evaluation.	-	-	-	-	-	-
DeGuard[4]	F-Droid apps, exclude obfuscated.	1.8K	-	-	-	-	A
JSNice[25, 26]	Github projects, exclude obfuscated.	10.6K	327.2K	-	-	-	A
JSNaughty[30]	Github projects, exclude obfuscated, file-level dedup.	100.0K	735.9K	-	-	-	P
Context2Name[3]	Files from JS150, exclude obfuscated.	-	98.0K	-	-	-	A
Pigeon[1]	Github projects, file-level dedup.	26.6K	2.8M	-	-	-	A
JSNeat[29]	Github projects, exclude obfuscated, file-level dedup.	12.0K	322.0K	967.0K	-	-	A
DOBF[18]	Github projects, file-level dedup.	-	-	-	-	-	A,P,R,F1
Seq2Seq[28]	Files from JS150, file-level dedup.	-	98.0K	-	-	-	A

5.3 Artifact Evaluation

Table 6. Experience of Artifact Evaluation

Literature	Experimental Discoveries
Debin	1. Debin provides an online tool [10], but it returned a “502 Bad Gateway” error. 2. Debin’s performance drops clearly on our self-constructed dataset. 3. Most correctly recovered names are single characters in the test file provided in the repository. 4. Debin prefers to predict long names.
DIRE	1. Replication results match those reported in the paper. 2. DIRE’s performance drops significantly on our self-constructed dataset.
Nero	1. Replication results match those reported in the paper. 2. We failed to run the comparison evaluation because Nero depends on IDA Pro, which no longer supports Python 2.
Punstrip	We failed to run the artifact. As noted in the README, “Some scripts may be broken, this is a partial release with some latest research advancements held back.”
DIRECT	1. Replication results match those reported in the paper. 2. DIRECT uses the exact dataset from DIRE and does not include any data preprocessing steps. 3. DIRECT’s performance drops significantly on our self-constructed dataset.
Artuso et al.	1. The evaluation results match those presented in the paper. 2. The model’s performance drops significantly on our self-constructed dataset. 3. The implementation quality is low and differs from what is described in the paper. 4. The model prefers to predict short names.
SymLM	We failed to run the artifact as the trained model provided is not matched to code.

(Continued)

Table 6. Continued

Literature	Experimental Discoveries
DIRTY	1. DIRTY provides an online tool [7], but it only shows pre-stored examples. 2. Replication results match those reported in the paper. 3. DIRTY's performance drops clearly on our self-constructed dataset.
XFL	1. XFL's performance drops clearly on our self-constructed dataset. 2. XFL predicts only compound-word names, with the default top five most probable subtokens.
Cfg2Vec	1. We failed to run the artifact as the provided model download link is invalid. 2. The data preprocessing code is incomplete.
AsmDepictor	1. Replication results match those presented in the paper. 2. We failed to run the comparison evaluation because the data preprocessing code is incomplete.
BinAdapter	We failed to run the artifact because the authors did not release the data and trained model.
Epitome	We failed to run the artifact because it references an undefined attribute and lacks required files.
NER	1. Replication results match those presented in the paper. 2. NER's performance drops clearly on our self-constructed dataset.
VarBERT	1. Replication results match those presented in the paper. 2. VarBERT's performance drops clearly on our self-constructed dataset. 3. VarBERT produces very few examples because Joern often fails to analyze C-like decompiled code.
ReSym	1. Replication results match those presented in the paper. 2. ReSym's performance drops noticeably on our self-constructed dataset.
SymGen	1. Replication results are notably better than those reported in the paper. 2. SymGen's performance drops slightly on our self-constructed dataset.
GENNM	1. Replication results match those presented in the paper. 2. GENNM's performance drops slightly on our self-constructed dataset.
LLASM	We failed to run the artifact because the authors did not release the dataset or the preprocessing procedure.
Duboue	We failed to run the artifact because the authors did not release the necessary resources, such as installation files, data, trained model, or scripts documentation.
DeMinify	We failed to run the artifact, as it only provides a project list and lacks data preparation instructions.
DeGuard	We failed to run the artifact, as the online tool consistently errors when processing APK files, including its own sample.
JSNice	1. Replication results match those presented in the paper. 2. JSNice supports only JavaScript ES5, but JavaScript ES6 is now more commonly used.
IDA Pro	1. IDA Pro executes efficiently and is easy to deploy. 2. IDA Pro's symbol name recovery performance is worse than dedicated approaches.
GPT-4o	1. GPT-4o executes efficiently and is easy to deploy. 2. GPT-4o's symbol name recovery performance is worse than Claude Haiku and dedicated approaches.
Claude Haiku	1. Claude Haiku executes efficiently and is easy to deploy. 2. Claude Haiku achieves better name recovery performance than GPT-4o, but with much higher token cost.

5.1.1 Comparison Study.

For approaches/models that use native resources (i.e., CPU), we use time overhead as the metric of computational cost. Specifically, we measured the end-to-end time overhead under the same hardware, i.e., from the original binary to the final recovered names.

Table 10. Time for all evaluated artifacts

Artifact	IDA Pro	NER	XFL	DIRE	Debin	DIRTY	Artuso et al.	VarBERT	GENNM	DIRECT	ReSym	SymGen
Time(mins)	10	13	13	16	32	63	96	100	290	428	1,280	2,280

The time overhead for all evaluated artifacts (from shortest to longest) is shown in Table 10. However, we argue that they are listed for reference purposes rather than comparative purposes because the time overhead is affected by many implementation-level factors, for example: (1) The artifacts differ in whether their preprocessing pipelines support multi-threading, which can accelerate preprocessing. (2) The artifacts differ in their inference hyperparameter settings, such as batch size, where larger batch sizes accelerate inference. (3) The artifacts differ in whether they support GPU acceleration, which can substantially accelerate inference.

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